Nvidia GPU Microarchitecures

Siddhant Gedam
Project Associate
C-DAC Bengaluru

Source : NVIDIA

Nvidia GPU microarchitecures

- *Tesla*
- *Fermi*
- *Kepler*
- *Maxwell*
- *Pascal*
- *Volta*
- *Turing*
- *Ampere*

GPU


PCI Express 3.0 Host Interface
GigaThread Engine
Memory Controller
Memory Controller
Memory Controller
Memory Controller
Memory Controller
Memory Controller
SMX
SMX
SMX
SMX
SMX
SMX
SMX
SMX
L2 Cache
SMX
SMX
SMX
SMX
SMX
SMX
SMX

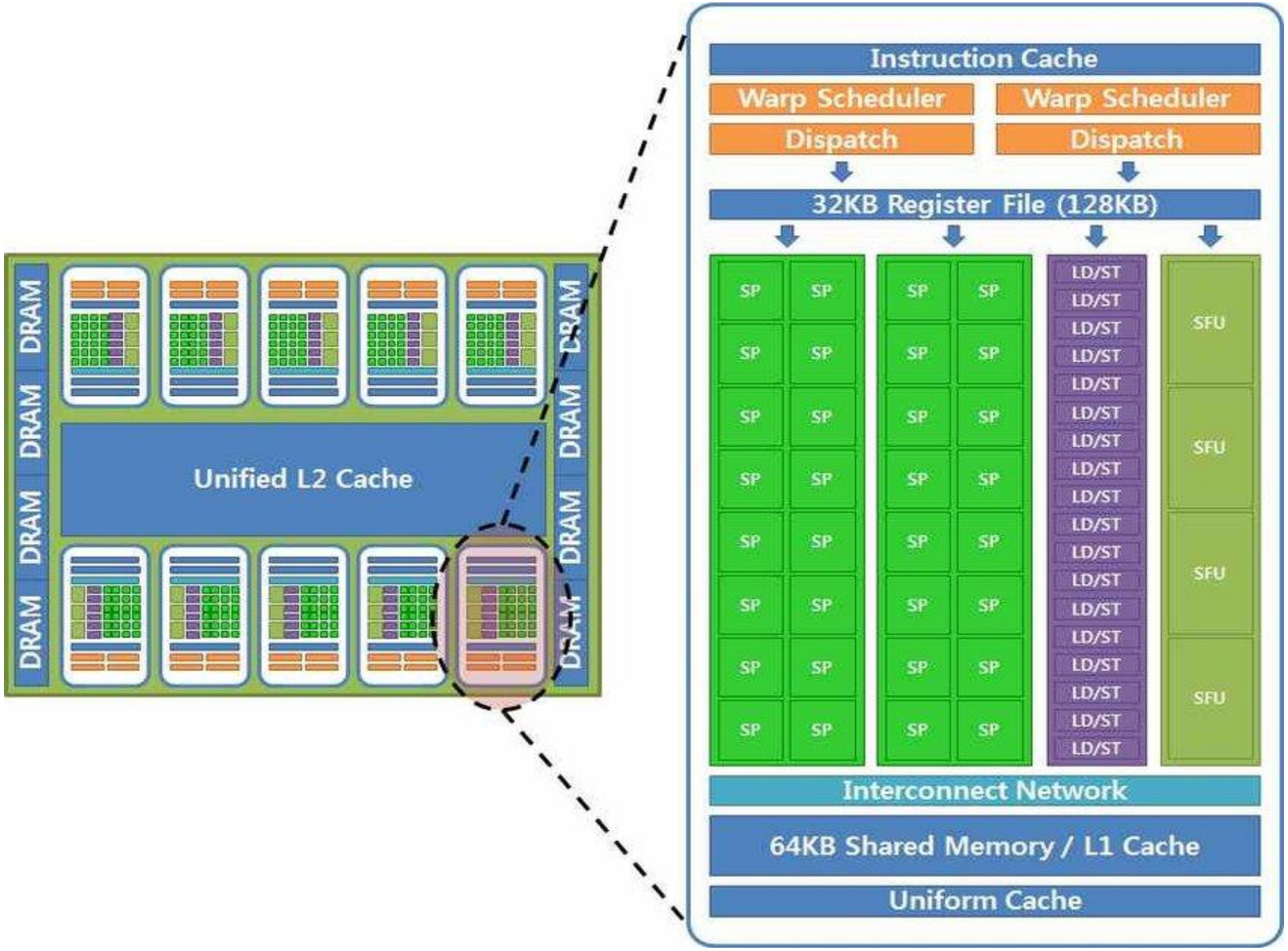
Streaming Multiprocessor(SM)

- Basic building block
- Contains

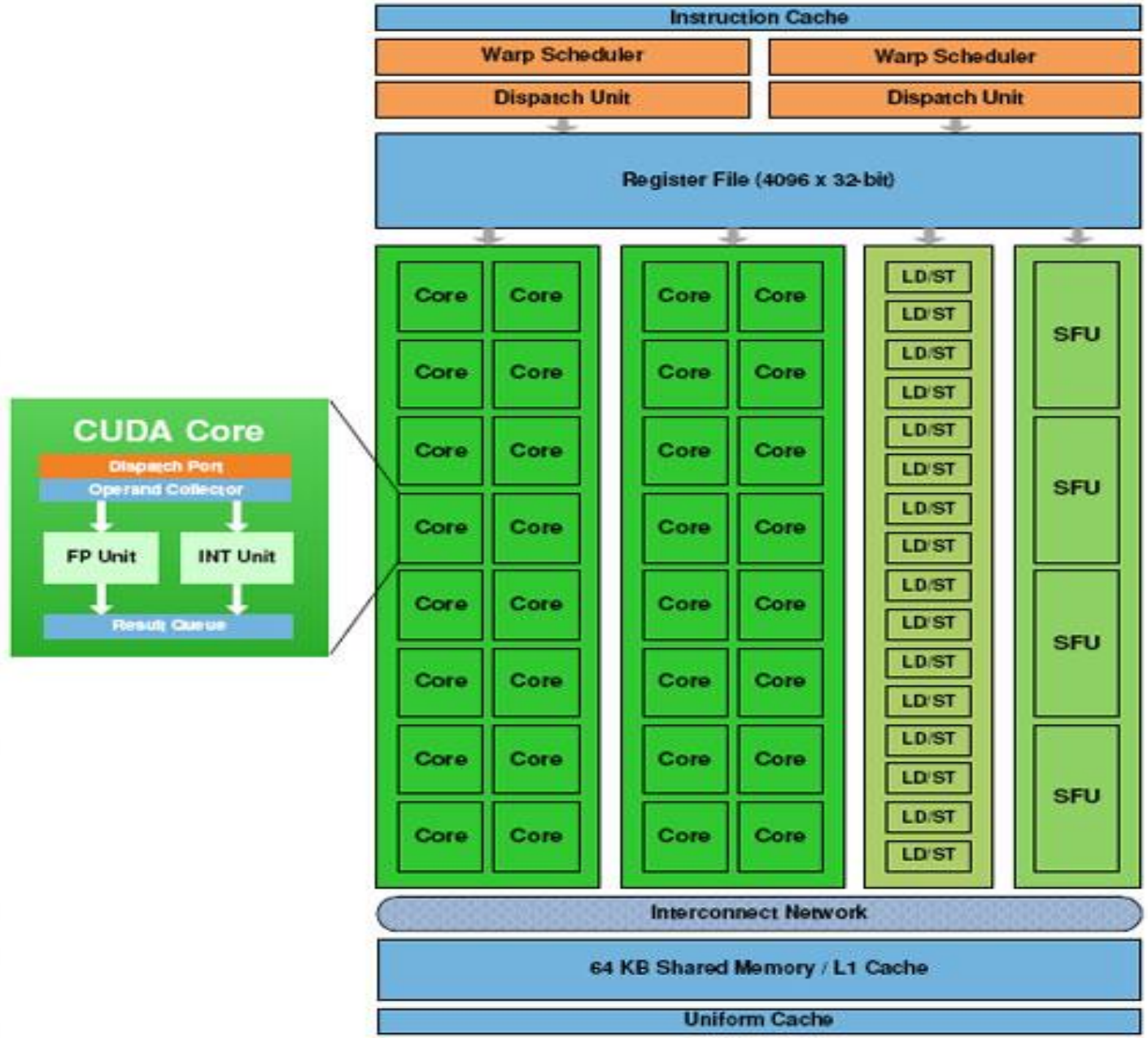
– Multiple CUDA cores
– Special Function Units (SFUs)
– Register file
– L1 cache / Shared memory
– Massive Parallelism
–….

Fermi GPU

The GF100 SM has 32 CUDA cores, four times the number per SM in the GT200. Each GPC includes four SMs, for a total of 512 CUDA cores in the GF100. Each SM includes a configurable cache or local memory for a total of 48 Kbytes of shared memory with 16 Kbytes of L1 cache. This memory can be reconfigured to use 16 Kbytes of shared memory and 48 Kbytes of L1 cache. There was no L1 cache for the SM in the GT200 architecture. Figure 3 shows a single SM with 32 CUDA cores.



Fermi GPU


Instruction Cache
Warp Scheduler
Warp Scheduler
Dispatch Unit
Dispatch Unit
Register File (4096 x 32-bit)
Core
LD/ST
SFU
Interconnect Network
64 KB Shared Memory / L1 Cache
Uniform Cache
CUDA Core
Dispatch Port
Operand Collector
FP Unit
INT Unit
Result Queue

Kepler

- 192 single-precision CUDA cores
- 64 double-precision units
- 32 special function units
- 32 load/store units
- 4 warp schedulers
- 8 instruction dispatch units
- 48 KB Read-Only Data Cache
- 64/128 KB L1 cache / shared memory



Pascal

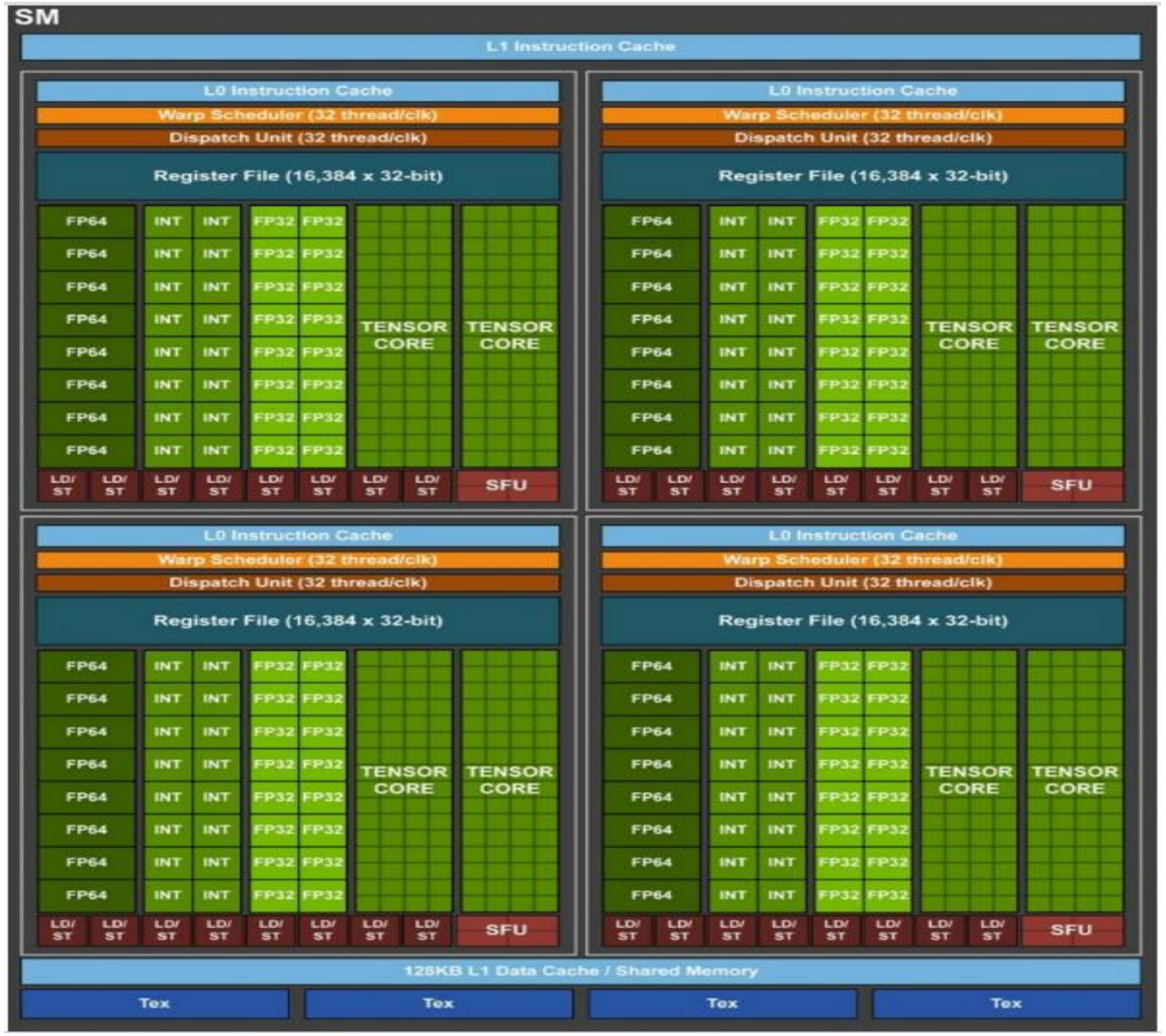
- two processing blocks, each with

– 32 single-precision CUDA Cores
– 16 double precision (FP64) CUDA Cores
– one instruction buffer
– one warp scheduler
– two dispatch units



Volta

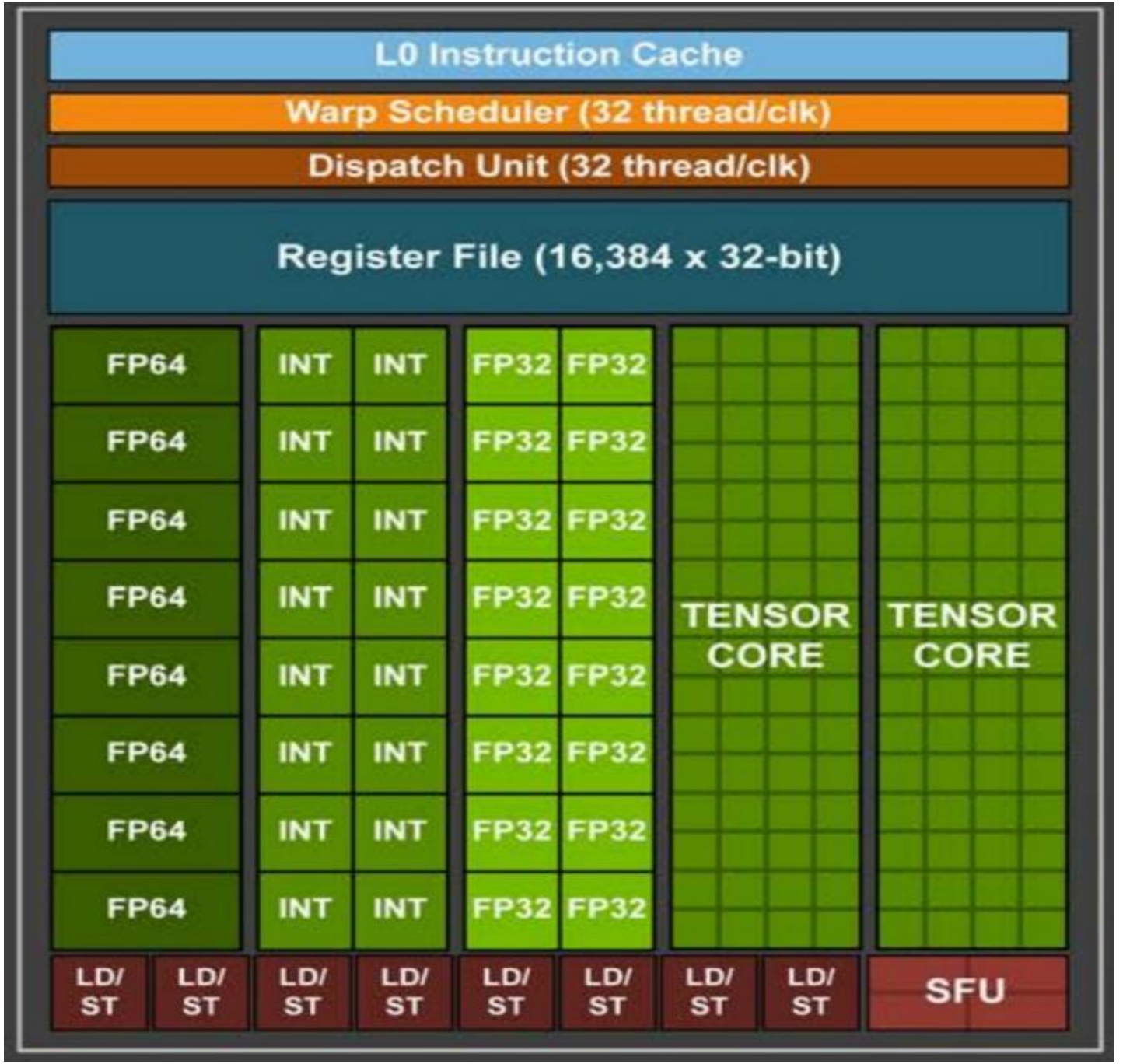
Four processing blocks



Volta *(contd)*

each processing block has

- 16 FP32 Cores
- 8 FP64 Cores
- 16 INT32 Cores
- 2 new mixed-precision Tensor Cores
- one L0 instruction cache
- one warp scheduler
- one dispatch unit
- 64 KB Register File



Ampere

each processing block has

- 32 FP32 Cores
- 16 FP64 Cores
- 64 INT32 Cores
- Tens of Tensor Cores
- one L0 instruction cache
- one warp scheduler
- one dispatch unit
- 128 KB Register File



New GPUs Architecture launch by NVIDIA

- **HOPPER –H100 (Grace Hopper)**
- **ADA LOVELACE**

Hopper

each processing block
has

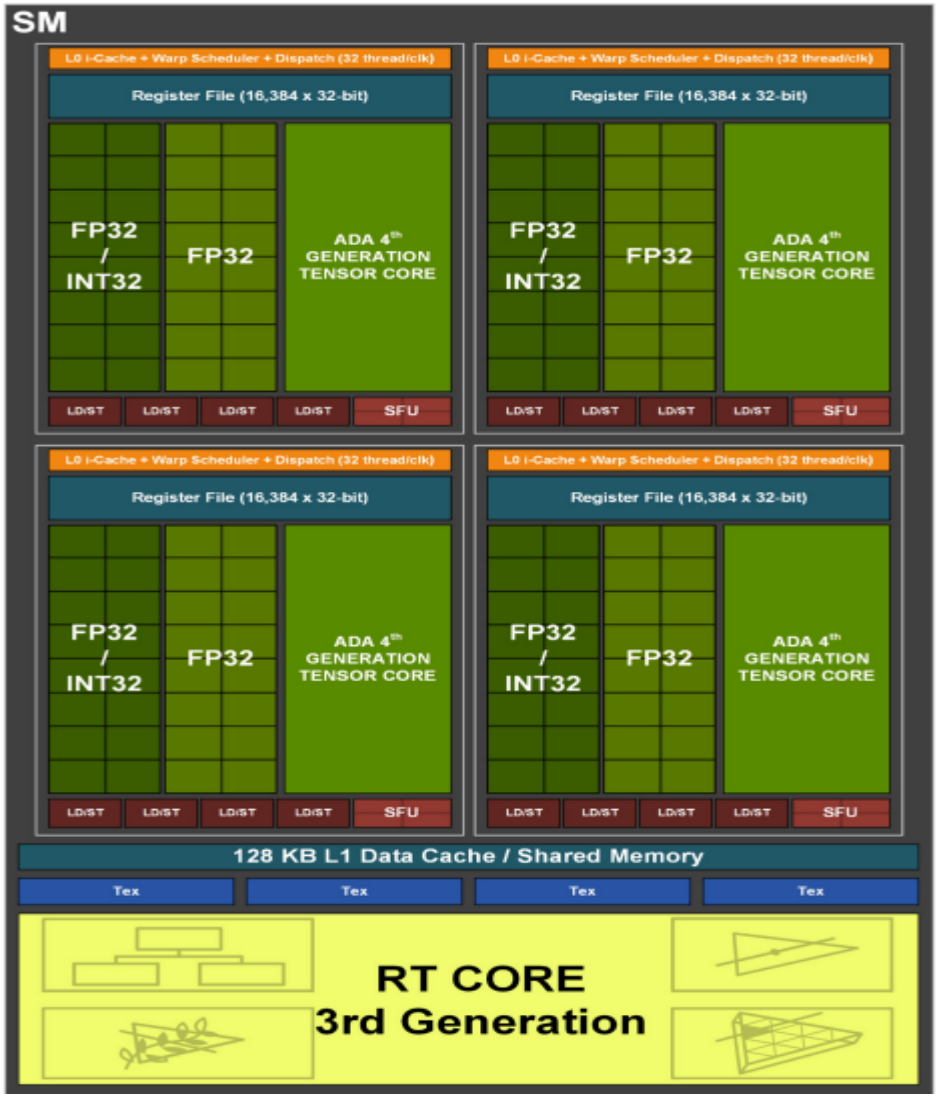
- 128 FP32 Cores
- 64 FP64 Cores
- 64 INT32 Cores
- 4 4th-generation Tensor Cores per SM, 456 per GPU
- 60 MB L2 cache
- one warp scheduler
- one dispatch unit
- 264 KB Register File



ADA LOVELACE

each processing block has

- ❑ 128 FP32 Cores
- ❑ 64 INT32 Cores
- ❑ 4 Ada 4th-generation Tensor Cores per SM
- ❑ 128 L1 cache
- ❑ one warp scheduler
- ❑ one dispatch unit
- ❑ 264 KB Register File



Question :
Check the improvement for each architecture and different architecture (Generation) ?